

Scalarizing functions for selecting a solution $x \in X_{WE}$

Definitions:

1. A **Multi-Objective Optimization** problem:

$$\begin{aligned} &\text{minimize} \quad f(x) = [f_1(x), \dots, f_d(x)] \\ &\text{subject to} \quad x \in X \end{aligned}$$

2. The **Weighted Tchebychev** Scalarizing Function $s_\lambda(X, \lambda, r^*)$:

$$\begin{aligned} &\min \left[\max_{k=1, \dots, d} \{ \lambda_k (r_k^* - f_k(x)) \} \right] \\ &\text{subject to} \quad x \in X \end{aligned}$$

- Output: $x^* \in X_{WE}$

3. The **Augmented Weighted Tchebychev** Scalarizing Function $s_\lambda^\rho(X, \lambda, r^*, \rho)$:

$$\begin{aligned} &\min \left[\max_{k=1, \dots, d} \{ \lambda_k (r_k^* - f_k(x)) \} + \rho \sum_{k=1}^d (r_k^* - f_k(x)) \right] \\ &\text{subject to} \quad x \in X \end{aligned}$$

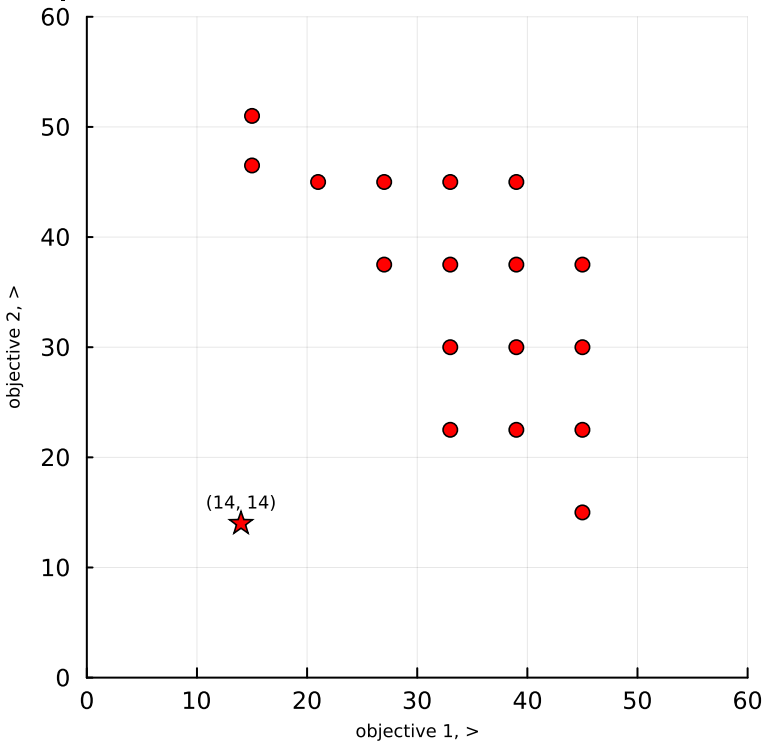
- Output: $x^* \in X_E \subseteq X_{WE}$

Example:

```
z(X) composed of 17 points
zUtopian = (14 , 14)
```

trace_Tcheby (generic function with 1 method)

W|AW Tchebychev Scalarizing Function



Weight (λ_1, λ_2) :

0.51

☐ `Weights: ( 0.510 , 0.490 )`

Value of the equidistance curve:

☐ Lsimple ☐ Laugmented

Show y^* : ☐

☐ `(iy_W, d_W, X[iy_W], Y[iy_W]) = (6, 9.31, 33, 30.0)
[18.87, 16.575, 15.81, 11.985, 15.81, 9.31, 9.31, 11.985, 15.81, 12.25, 12.25,
12.25, 15.81, 15.19, 15.19, 15.19, 15.19]
(iy_WA, d_WA, X[iy_WA], Y[iy_WA]) = (7, 9.585, 33, 22.5)
[19.25, 16.91, 16.19, 12.35, 16.25, 9.66, 9.585, 12.41, 16.310000000000002, 12
.585, 12.66, 12.735, 16.37, 15.51, 15.584999999999999, 15.66, 15.735]`